Point convergence of IREG-PRM$^+$ without equilibrium uniqueness

Abstract

For the strict current-state shifted-extragradient IREG-PRM$^+$ recurrence in a finite zero-sum matrix game, we prove that the entire strategy sequence converges to one Nash equilibrium, even when the equilibrium is not unique. Li and Huang already establish bounded regret norms and vanishing last-iterate gaps, but leave possible drift within the equilibrium set unresolved. The additional argument uses the existing regret accounting geometry: pairing the cumulative regret states with any fixed equilibrium gives a bounded nondecreasing scalar. At an equilibrium cluster point this scalar attains its limiting Cauchy–Schwarz bound, which forces the whole state sequence to converge. We specify the zero-state convention and include a self-contained account of the global ingredients. The result covers boundary equilibria and never-positive regret states. It assumes no strict limiting spectral condition and gives no rate.

Keywords: IREG-PRM$^+$; regret matching; zero-sum matrix games; Nash equilibrium; last-iterate convergence; nonunique equilibria

2020 Mathematics Subject Classification. Primary 91A26; Secondary 91A05, 91A68.

1 Introduction

Scale-invariant regret matching uses the cumulative regret norm to determine its effective step size. Zhang, Anagnostides, and Sandholm [2] introduce the scale-invariant construction and analyze its regret and convergence properties. For its extragradient form, IREG-PRM$^+$, Li and Huang [1] prove global norm saturation and vanishing last-iterate Nash gaps. Their global analysis gives point convergence when the equilibrium is unique; their local linear theory has additional hypotheses. Appendix B.4 of [1] explicitly leaves open whether a trajectory can drift along a nonunique equilibrium set.

This note resolves that point-selection question for the literal strict recurrence of [1, Algorithm 1 and Appendix B.1], initialized at zero with the uniform zero-state action. The contribution is the final selection argument in Section 5. We do not claim norm saturation, the scalar shift identity, or the zero-sum regret accounting inequalities as new. Sections 3–4 give elementary versions of those established ingredients to make all zero-norm cases explicit. In particular, the proof does not obtain convergence by imposing a spectral contraction condition or identifying a unique equilibrium.

2 The precise recurrence

Let $A \in \mathbb{R}^{n\times m}$, where $n, m \geq 1$. The row player chooses $x \in \Delta_n$ to maximize $x^{\mathsf{T}}Ay$, and the column player chooses $y \in \Delta_m$ to minimize it. Write v^* for the game value. An

equilibrium (p, q) satisfies

$$p^{\mathsf{T}} A y \geq v^* \geq x^{\mathsf{T}} A q \qquad (x \in \Delta_n,\ y \in \Delta_m). \tag{1}$$

The Euclidean norm is denoted by $\|\cdot\|$, the ℓ_1 norm by $\|\cdot\|_1$, and positive-part clipping by $P(w) = [w]_+$.

The nonnegative cumulative states are $a_t \in \mathbb{R}^n$ and $b_t \in \mathbb{R}^m$, with $a_1 = b_1 = 0$. Define the pre-strategies by

$$\widetilde{x}_t = \begin{cases} a_t/\|a_t\|_1, & a_t \neq 0, \\ \mathbf{1}/n, & a_t = 0, \end{cases} \qquad \widetilde{y}_t = \begin{cases} b_t/\|b_t\|_1, & b_t \neq 0, \\ \mathbf{1}/m, & b_t = 0. \end{cases} \tag{2}$$

Both pre-strategies are formed before either prediction. If $a_t \neq 0$, set $m_{1t} = A\widetilde{y}_t$ and choose the unique shift γ_{1t} satisfying

$$r_{1t} = a_t + m_{1t} - \gamma_{1t}\mathbf{1}, \qquad s_{1t} = P(r_{1t}), \qquad \|s_{1t}\| = \|a_t\|, \qquad x_t = \frac{s_{1t}}{\|s_{1t}\|_1}. \tag{3}$$

For $b_t \neq 0$, the symmetric construction uses $m_{2t} = -A^{\mathsf{T}}\widetilde{x}_t$ and gives $r_{2t}, s_{2t}, \gamma_{2t}, y_t$. At a zero state, set its m_{it}, r_{it}, s_{it} to zero and its actual strategy to the uniform pre-strategy; no shift or zero-denominator normalization is performed.

Once both actual strategies have been formed, let $x_{1t} = x_t$, $x_{2t} = y_t$, and set

$$\begin{aligned} u_{1t} &= Ay_t, & u_{2t} &= -A^{\mathsf{T}}x_t, \\ g_{it} &= u_{it} - m_{it} - \langle u_{it} - m_{it}, x_{it}\rangle\mathbf{1}, & i &\in \{1, 2\}, \\ a_{t+1} &= P(r_{1t} + g_{1t}), & b_{t+1} &= P(r_{2t} + g_{2t}). \end{aligned} \tag{4}$$

The predictor r_{it} in the final update is *unclipped*. Neither a lagged prediction nor the alternative update $P(s_{it} + g_{it})$ is used.

Remark (Source convention). The zero-state branch of [1, Algorithm 1] assigns the carried-forward default strategy but directs the reader to its generic normalization line, which would divide zero by zero. We preserve the assigned strategy and proceed to observation and correction, as specified in its Appendix B.1. The default is uniform. Lemma 1 below proves that a norm cannot return to zero after becoming positive; consequently the carried-forward default equals the uniform strategy on every zero-state round. This states explicitly the branch interpretation used in the theorem.

Theorem 1. *For every finite real matrix A, the recurrence (2)–(4) from $a_1 = b_1 = 0$ is well-defined and there is a Nash equilibrium (p, q) such that*

$$(x_t, y_t) \longrightarrow (p, q), \qquad (\widetilde{x}_t, \widetilde{y}_t) \longrightarrow (p, q).$$

The cumulative states also converge. No uniqueness, interiority, strict complementarity, nondegeneracy, or strict limiting spectral condition is required.

3 Shift geometry and equilibrium accounting

For a positive state, existence and uniqueness of the shift follow because $\gamma \mapsto \|P(a + m - \gamma\mathbf{1})\|$ is continuous, strictly decreasing while positive, unbounded as $\gamma \to -\infty$, and zero for $\gamma \geq \max_j(a_j + m_j)$. These facts also imply that every finite round is well-defined.

Lemma 1. *Consider one positive state $a \in \mathbb{R}^d_{\geq 0}$, with prediction m, shift γ, predictor $r = a + m - \gamma\mathbf{1}$, $s = P(r)$, and $x = s/\|s\|_1$. Put $N = \|a\| = \|s\| > 0$, $L = \|s\|_1$, and $\beta = \langle m, x\rangle - \gamma$. Then*

$$2L\beta = \|s - a\|^2, \qquad \beta \geq 0. \tag{5}$$

For any centered correction $g = (u - m) - \langle u - m, x\rangle\mathbf{1}$,

$$N^2 \leq \|P(r + g)\|^2 \leq N^2 + \|g\|^2. \tag{6}$$

If $\widetilde{x} = a/\|a\|_1$ and $C_d = 2\sqrt{d}(1 + \sqrt{d})^2$, then

$$\|x - \widetilde{x}\|^2 \leq C_d\beta/N. \tag{7}$$

Proof. Where $s_j > 0$, one has $r_j = s_j$. Hence

$$L\beta = \langle s, m - \gamma\mathbf{1}\rangle = \langle s, r - a\rangle = N^2 - \langle s, a\rangle = \tfrac{1}{2}\|s - a\|^2.$$

Also $\langle g, s\rangle = 0$ and $P(r + g) \geq r + g$ coordinatewise, so

$$\langle s, P(r + g)\rangle \geq \langle s, r + g\rangle = N^2.$$

Cauchy–Schwarz proves the lower bound in (6). For the upper bound, $r \leq s$ gives $0 \leq P(r + g) \leq P(s + g)$, whence

$$\|P(r + g)\|^2 \leq \|s + g\|^2 = N^2 + \|g\|^2.$$

Finally, $\|a\|_1, \|s\|_1 \geq N$ and the normalization estimate give

$$\|x - \widetilde{x}\| \leq \frac{1 + \sqrt{d}}{N}\|s - a\|.$$

Use (5) and $L \leq \sqrt{d}\,N$ to obtain (7). ∎

The identity (5) and the associated nonnegative shift term are part of the existing geometry in [1, Appendix A.1]; the constants above are chosen only for a short self-contained estimate. Equation (6) proves that each regret norm is nondecreasing, including transitions from zero. Define

$$N_1(t) = \|a_t\|, \qquad N_2(t) = \|b_t\|, \qquad \beta_{it} = \begin{cases} \langle m_{it}, x_{it}\rangle - \gamma_{it}, & N_i(t) > 0, \\ 0, & N_i(t) = 0. \end{cases}$$

Lemma 2. *For every equilibrium (p, q), the scalar*

$$H_t(p, q) = \langle p, a_t \rangle + \langle q, b_t \rangle$$

satisfies

$$H_{t+1}(p, q) - H_t(p, q) \geq \beta_{1t} + \beta_{2t} \geq 0. \tag{8}$$

Moreover $0 \leq H_t(p, q) \leq \|p\| N_1(t) + \|q\| N_2(t)$.

Proof. Writing $v_t = x_t^{\mathsf{T}} A y_t$, expansion of (4) gives

$$\begin{aligned} a_{t+1} &= P(a_t + A y_t - v_t \mathbf{1} + \beta_{1t} \mathbf{1}), \\ b_{t+1} &= P(b_t - A^{\mathsf{T}} x_t + v_t \mathbf{1} + \beta_{2t} \mathbf{1}). \end{aligned} \tag{9}$$

At a zero state the same formulas follow directly from $r_{it} = m_{it} = 0$ and $\beta_{it} = 0$, without a shift. Pairing with nonnegative probability vectors p, q, and using $P(w) \geq w$, yields

$$H_{t+1} - H_t \geq p^{\mathsf{T}} A y_t - x_t^{\mathsf{T}} A q + \beta_{1t} + \beta_{2t}.$$

Now use (1); the upper bound follows from Cauchy–Schwarz. ∎

This pairing retains a fixed equilibrium comparator in the standard zero-sum regret accounting argument; see [1, Appendix A.2] and [2]. The selection proof will use its limiting value, rather than a new regret estimate.

4 Global ingredients and zero states

Li and Huang [1, Theorems 1–2] establish bounded norms and vanishing last-iterate gaps. We record a proof of the ingredients needed here so that the positive and zero branches are covered within one recurrence.

Lemma 3. *The limits $N_{i,\infty} = \lim_t N_i(t)$ exist and are finite. Moreover,*

$$\sum_{t=1}^{\infty} (\beta_{1t} + \beta_{2t}) < \infty, \qquad \|x_t - \widetilde{x}_t\| + \|y_t - \widetilde{y}_t\| \longrightarrow 0. \tag{10}$$

Every cluster point of (x_t, y_t) is a Nash equilibrium.

Proof. *Boundedness when both norms become positive.* Choose a finite τ after both have become positive. Set

$$\alpha_1 = \frac{N_1(\tau)}{C_n} > 0, \quad \alpha_2 = \frac{N_2(\tau)}{C_m} > 0, \quad K_1 = (1 + \sqrt{n}) \|A\|_{\mathrm{op}}, \quad K_2 = (1 + \sqrt{m}) \|A\|_{\mathrm{op}}.$$

For $t \geq \tau$, the current-state predictions imply

$$\|g_{1t}\| \leq K_1 \|y_t - \widetilde{y}_t\|, \qquad \|g_{2t}\| \leq K_2 \|x_t - \widetilde{x}_t\|. \tag{11}$$

For example, $g_{1t} = (I - \mathbf{1}x_t^{\mathsf{T}})A(y_t - \widetilde{y}_t)$ and $\|I - \mathbf{1}x_t^{\mathsf{T}}\|_{\mathrm{op}} \le 1 + \sqrt{n}$. By norm monotonicity and (7), the two squared strategy discrepancies are at most β_{1t}/α_1 and β_{2t}/α_2. Let

$$Q_t = N_1(t)^2 + N_2(t)^2, \qquad M = \max\{K_1^2/\alpha_2, K_2^2/\alpha_1\}.$$

Equation (6) now gives $Q_{t+1} - Q_t \le M(\beta_{1t} + \beta_{2t})$. Fix one equilibrium, whose existence follows from finite minimax. For $T \ge \tau$, Lemma 2 gives

$$Q_T \le Q_\tau + M(H_T - H_\tau) \le Q_\tau + M\{N_1(T) + N_2(T)\} \le Q_\tau + M\sqrt{2Q_T}. \tag{12}$$

Thus $\sqrt{Q_T} \le M/\sqrt{2} + \sqrt{M^2/2 + Q_\tau}$. Each nondecreasing norm has a finite positive limit.

Boundedness in the remaining cases. If a player never has a positive norm, its actual and pre-strategies are the same uniform vector at every round. If only the column player becomes positive, then from its first positive round onward $m_{2t} = u_{2t} = -A^{\mathsf{T}}(\mathbf{1}/n)$, so $g_{2t} = 0$. Norm preservation makes its norm constant. The case with the players exchanged is identical. If neither norm becomes positive, both are identically zero. These cases prove boundedness without a positive lower bound for a zero norm.

Vanishing prediction discrepancy. The now bounded monotone pairing and (8) imply the summability in (10). For an eventually positive player, (5) and $\|s_{it}\|_1 \le \sqrt{d_i}N_i(t)$, where $d_1 = n$ and $d_2 = m$, give

$$\|s_{1t} - a_t\| \longrightarrow 0, \qquad \|s_{2t} - b_t\| \longrightarrow 0, \tag{13}$$

with the corresponding equality identically zero for a never-positive player. Equation (7), using a positive lower norm after activation, proves the remaining assertion in (10); zero-state strategies already coincide.

Equilibrium cluster points. Take a subsequence on which $(x_t, y_t) \to (p, q)$. Compactness of the simplices guarantees such a subsequence, and (10) gives the same limit for the pre-strategies. If $N_{1,\infty} > 0$, then along this subsequence

$$a_t = N_1(t)\frac{\widetilde{x}_t}{\|\widetilde{x}_t\|} \longrightarrow \bar{a} = N_{1,\infty}\frac{p}{\|p\|}, \qquad s_{1t} \longrightarrow \bar{a}. \tag{14}$$

The shift lies between $\min_j m_{1t,j}$ and $\max_j m_{1t,j}$: a shift above the latter strictly decreases the norm, while one below the former strictly increases it. These bounds are finite uniformly in t. Extracting a further subsequence with $\gamma_{1t} \to \bar{\gamma}$ and taking limits in (3) gives

$$\bar{a} = P(\bar{a} + Aq - \bar{\gamma}\mathbf{1}). \tag{15}$$

For $p_j > 0$ this says $(Aq)_j = \bar{\gamma}$, and for $p_j = 0$ it says $(Aq)_j \le \bar{\gamma}$. Thus p is a best response to q.

If $N_{1,\infty} = 0$, norm monotonicity implies $a_t = 0$ and $x_t = \mathbf{1}/n$ for every t. The update gives

$$0 = P\Big(Ay_t - (x_t^{\mathsf{T}}Ay_t)\mathbf{1}\Big).$$

Every coordinate of Ay_t is at most its uniform average, so all these coordinates are equal. Passing to the limit again makes $p = \mathbf{1}/n$ a best response to q. The symmetric argument with $-A^\mathsf{T}$ proves that q is a best response to p, including its never-positive case. Hence (p, q) is Nash. ∎

5 Selection of a single equilibrium

Proof of Theorem 1. Choose one strategy cluster point (p, q). By Lemma 3, it is an equilibrium. Along its defining subsequence, (14) and its column counterpart give

$$a_t \longrightarrow N_{1,\infty}\frac{p}{\|p\|}, \qquad b_t \longrightarrow N_{2,\infty}\frac{q}{\|q\|}. \tag{16}$$

The same formulas hold for a never-positive player because its state and limiting norm are zero. Every simplex vector has nonzero Euclidean norm, so the displayed denominators are always positive.

For this particular equilibrium, Lemma 2 makes $H_t(p, q)$ bounded and nondecreasing. Its full-sequence limit can therefore be evaluated on the subsequence in (16):

$$\lim_t H_t(p, q) = N_{1,\infty}\|p\| + N_{2,\infty}\|q\|. \tag{17}$$

Consider the two nonnegative Cauchy–Schwarz deficits

$$D_{1t} = N_1(t)\|p\| - \langle p, a_t\rangle, \qquad D_{2t} = N_2(t)\|q\| - \langle q, b_t\rangle.$$

By norm convergence and (17), $D_{1t}+D_{2t} \to 0$. Hence both deficits tend to zero separately. The exact identity

$$\left\| a_t - N_1(t)\frac{p}{\|p\|}\right\|^2 = \frac{2N_1(t)}{\|p\|}D_{1t} \tag{18}$$

and its column counterpart, together with boundedness of the norms, prove that (16) holds along the entire sequence.

If $N_{1,\infty} > 0$, normalization gives $\widetilde{x}_t \to p$, and (10) gives $x_t \to p$. If $N_{1,\infty} = 0$, both strategies are identically uniform and therefore equal to their cluster limit p. The same alternatives apply to the column player. This proves all the conclusions. ∎

The proof includes nonunique and boundary equilibrium sets, rectangular matrices, duplicated actions, one-action players, and the zero matrix. It requires neither support freezing nor summable first-order path length. It gives no rate and does not establish a strict limiting local spectral inequality.

References

[1] B. Li and L. Huang. Why Last-Iterate Scale-Invariant Regret Matching Converges Linearly? *arXiv preprint*, arXiv:2609.38162v1, 2026.

[2] B. H. Zhang, I. Anagnostides, and T. Sandholm. Scale-invariant regret matching and online learning with optimal convergence: bridging theory and practice in zero-sum games. *arXiv preprint*, arXiv:2510.04407v1, 2025.